Anton Liutin

Department of Agricultural & Applied Economics, University of Wisconsin–Madison
304 Taylor Hall, 427 Lorch St, Madison, WI 53706
liutin@wisc.edu · +1 (608) 640-1506 · antonliutin.com

EDUCATION

PhD, Agricultural & Applied Economics 2021 – Expected May 2027
University of Wisconsin–Madison — dissertator. Advisor: Paul Castañeda Dower

MA, Economics 2020
New Economic School, Moscow, Russia
Thesis: "Polygamy and Inequality in a Male-Dominant Society" (advisor: Hosny Zoabi)

BS, Theoretical Physics 2018
National Research Nuclear University MEPhI (Moscow Engineering Physics Institute), Moscow, Russia

RESEARCH INTERESTS

Fields: Water and natural-resource economics · development economics · institutional and political economy · behavioral and experimental economics · economic history

Methods: Machine learning and text-as-data · spatial econometrics and remote sensing · network and graph methods · field and framed experiments

JOB MARKET PAPER

"Who Runs Short in a Drought? Water Rights and Canal Delivery in Arizona"
Full working draft. Links 28,588 Arizona fields to a reconstructed surface-water network over 2016–2023. In junior-rights irrigation districts, which lose surface water first during sustained drought, fields with more farmed land upstream are left fallow more often and show larger falls in crop yield and water use, even holding each field's water rights, well access, crop, and local weather fixed; farmed land downstream shows no such link. The loss is worth about 9% of these fields' net return in drought years. There is no such loss in senior-rights districts or in the more severe 2010–2014 drought, when deliveries were not cut. Monthly models that route canal supply and crop demand through each district's network estimate how each district shares water, matching its documented delivery rules; fields without a well use less water when less reaches them. Latest version: antonliutin.com/jmp.pdf.

WORKING PAPERS

"Behavioral Games and Water-Saving Technology Adoption in Uzbekistan" (with Jamshid Shukurullaev)
Working draft. Framed field experiment with 190 farmers across four regions, drawing participants from the IFPRI farmer survey; conducted with CGIAR SPIA support. Latest version: antonliutin.com/behavioral-games.pdf.

"Double Vision: Donor Information, Farmer Voice, and Irrigation Policy Reform in Uzbekistan" (with Muzna Alvi, Paul Castañeda Dower, and Claudia Ringler)
Sets about 330,000 farmer Telegram messages against media coverage, donor documents, and legislation to ask whose concerns the law follows. Four state newspapers OCR'd in house; six channels classified in two languages by an ensemble of language models, checked against an expert-coded sample. IFPRI collaboration. Latest version: antonliutin.com/double-vision.pdf.

"Water Conservation Through Simplified Drip Irrigation Scheduling for Cotton Cultivation in Uzbekistan" (Jamshid Shukurullaev, Ahmad Hamidov, and **Anton Liutin**)
Complete manuscript. Three-season paired field validation plus a 32-farmer adoption panel (difference-in-differences).

WORK IN PROGRESS

"Impact Evaluation of Drip Irrigation in Uzbekistan" (with Paul Castañeda Dower, Muzna Alvi, and Claudia Ringler)
Survey-based impact evaluation of drip-irrigation adoption using the 2024 farmer survey; IFPRI / CGIAR NEXUS Gains collaboration.

Community testimony and environmental change in Alaska (with Paul Castañeda Dower)
Language-model analysis of over a million turns of hearing testimony and regulatory comment from Alaska communities, paired with satellite measurements of environmental change.

PRESENTATIONS

New Economic School Brown Bag Seminar (online) April 2026
"From Position to Exposure: Drought Amplification in Directed Water Networks"

Central Asian Economics Conference, Astana, Kazakhstan August 2025
"Using Behavioral Games to Understand Water-Saving Technology Adoption in Uzbekistan"

CGIAR SPIA Webinar (online) August 2024
"Behavioral Games to Explore Water-Saving Technology Adoption in Uzbekistan"

NEXUS Gains / WEFE Central Asia Summer School, Tashkent, Uzbekistan August 2023
"Water-Saving Technologies in Uzbekistan"

Departmental seminars, University of Wisconsin–Madison 2022 – 2026
Regular presenter: DevLab, Environment and Resource Economics Seminar, and Student Research Colloquium

RESEARCH EXPERIENCE

Research Assistant, Paul Castañeda Dower Summer 2026
University of Wisconsin–Madison — community institutions and adaptation in Alaska: language-model analysis of hearing testimony and regulatory comment, paired with satellite measurements of environmental change

Research Assistant, Dustin Frye and Gisella Kagy Fall 2024 – Summer 2025
University of Wisconsin–Madison — environmental health and economic history

Research Assistant, Paul Castañeda Dower Summer 2023
University of Wisconsin–Madison — Russian economic history

Research Assistant, Corbett Grainger Summer 2022 – Spring 2023
University of Wisconsin–Madison — environmental and natural-resource economics

Research Assistant, Andrei Markevich Winter 2020 – Summer 2021
New Economic School — Russian economic history

Junior Researcher, Hosny Zoabi Winter 2020 – Summer 2021
New Economic School, Research Department — family and growth economics

Junior Researcher Winter 2017 – Summer 2018
P.N. Lebedev Physical Institute, Russian Academy of Sciences

FIELDWORK

Uzbekistan — framed field experiment on water-saving technology adoption Nov – Dec 2024
Designed and ran a lab-in-the-field experiment with 190 farmers across four regions, with Jamshid Shukurullaev; supported by CGIAR SPIA.

Uzbekistan — large-scale farmer survey (IFPRI / CGIAR NEXUS Gains) Sep – Nov 2024
Prepared and fielded a survey of small and large farmers across five regions; trained and monitored local enumerators.

Uzbekistan — qualitative fieldwork on water-saving technologies Summer 2023
Conducted focus-group discussions and in-depth farmer interviews across seven regions on the adoption of water-saving technologies, with Paul Castañeda Dower in collaboration with IFPRI.

TEACHING EXPERIENCE

Teaching Assistant, AAE 636: Applied Econometric Analysis I Fall 2026
University of Wisconsin–Madison — graduate applied econometrics

Co-Instructor, AAE Math Camp Summer 2026
University of Wisconsin–Madison — calculus and probability for incoming graduate students

Teaching Assistant, AAE 350: World Hunger and Malnutrition Spring 2024
University of Wisconsin–Madison — led four weekly discussion sections (cross-listed with Nutritional Sciences)

Teaching Assistant, AAE 215: Introduction to Agricultural and Applied Economics Fall 2023
University of Wisconsin–Madison — undergraduate introduction to agricultural economics

Teaching Assistant, Development Economics (MA) Fall 2020
New Economic School — course by Michele Valsecchi

Private Mathematics Tutor 2017 – 2022
Prepared students for mathematics-intensive economics BA/MA programs (New Economic School, HSE).

AWARDS AND FELLOWSHIPS

University Fellowship, UW–Madison 2021–22, 2025–26
Nine-month university fellowship supporting first-year doctoral study (2021–22) and fifth-year dissertation research (2025–26).

Deborah and David Penn Fund Research Presentation Award, UW–Madison ($1,000) 2025
AAE Graduate Committee award supporting presentation at the Central Asian Economics Conference, Astana.

CGIAR–SPIA Fellowship ($25,500) 2023
Research fellowship awarded following external peer review; supported the behavioral-games field experiment in Uzbekistan (NEXUS Gains collaboration).

CREECA Graduate Student Short-Term Research Award, UW–Madison ($3,000) 2023
Competitive award for the project "Current Cooperation Problems under Common-Pool Resources in Uzbekistan."

RESEARCH GRANTS AND PROJECTS

CGIAR GENDER Impact Platform, WEAI-Climate Learning Grant ($100,000) 2023–24
"Unveiling the Nexus: Climate-Smart Irrigation and Women's Empowerment in Uzbekistan." Named collaborator (survey design, fieldwork, analysis); PIs Muzna Alvi and Claudia Ringler.

SPIA Supplemental Funding, IFPRI / NEXUS Gains team ($40,950) 2024
Team funding for completion of survey, fieldwork, and outreach for the Uzbekistan research program; research-team member.

SPIA Scoping and Travel Grant ($9,300) 2023
Joint grant for scoping and preparatory fieldwork in Uzbekistan; co-applicant with Paul Castañeda Dower.

SKILLS AND LANGUAGES

Programming & software: Python (pandas, GeoPandas, NumPy, SciPy), Stata, R, ArcGIS, QGIS, LaTeX, Git

Machine learning & geospatial: Google Earth Engine (Sentinel-2, Landsat, OpenET); satellite and crop-sequence classification; directed-graph construction and validation at national scale

Text as data: LLM classification pipelines — in-house OCR (DocLayout-YOLO, Surya; Cyrillic), open-weight model ensembles with majority voting, validation against hand-coded gold standards

Languages: Russian (native), English (fluent)

REFERENCES

Paul Castañeda Dower (advisor), Associate Professor, Agricultural & Applied Economics, UW–Madison — pdower@wisc.edu
Corbett Grainger, Professor, Agricultural & Applied Economics, UW–Madison — corbett.grainger@wisc.edu
Daniel Phaneuf, Henry C. Taylor Professor, Agricultural & Applied Economics, UW–Madison — dphaneuf@wisc.edu